WORKED SOLUTION

STEP I 2010, Question 9

Mechanics • Friction

STEP I 2010, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/10-s1-q9

The key idea

Put the corner on the floor at a known place and walk to the centre of mass along the two perpendicular edges: that gives the distance from the wall. For the equilibrium, "limiting at both contacts" fixes each friction force as μ times its normal reaction, so the two resolutions give both reactions in terms of the weight. One moment equation then finishes the job, and writing $\mu = \tan\lambda$ turns the algebra into compound-angle formulae.

Solution

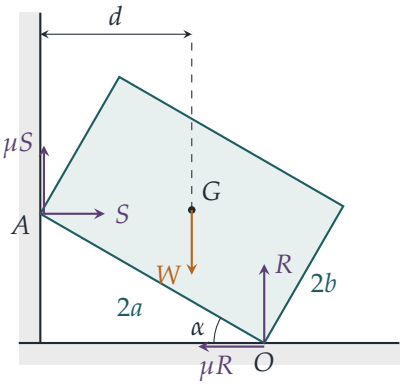
The model

The lamina is a rigid uniform rectangle of weight W, so its centre of mass G is its centre. Call O the corner on the floor and A the corner against the wall, so that OA is an edge of length $2a$ making angle α with the floor, and the edge of length $2b$ from O points up and away from the wall. The forces are:

- at O: normal reaction R upwards and friction F horizontally;
- at A: normal reaction S horizontally away from the wall and friction P vertically;
- the weight W downwards at G.

The lamina is on the point of slipping with O sliding away from the wall and A sliding down it, so friction opposes this: F acts towards the wall and P acts upwards. Since friction is limiting at both contacts,

$$F = \mu R, \qquad P = \mu S.$$



Distance of the centre of mass from the wall

Since OA has length $2a$ and makes angle α with the horizontal, O is a horizontal distance $2a\cos\alpha$ from the wall.

From O, reach G by going half-way along each of the two edges at O. Going a along OA (towards the wall) moves us $a\cos\alpha$ towards the wall. The edge of length $2b$ is perpendicular to OA, so it makes angle $\frac{1}{2}\pi - \alpha$ with the horizontal and leans away from the wall; going b along it moves us $b\cos(\frac{1}{2}\pi - \alpha) = b\sin\alpha$ away from the wall. Hence the distance of G from the wall is

$$d = 2a\cos\alpha - a\cos\alpha + b\sin\alpha = a\cos\alpha + b\sin\alpha,$$

as required. In the same way, G is a height $a\sin\alpha + b\cos\alpha$ above the floor, and A is a height $2a\sin\alpha$ above it.

The equilibrium condition

Resolving horizontally and vertically,

$$S = \mu R,$$
$$R + \mu S = W.$$

Substituting the first into the second gives $R(1+\mu^2) = W$, so

$$R = \frac{W}{1+\mu^2}, \qquad S = \frac{\mu W}{1+\mu^2}.$$

Now take moments about O, which removes R and μR. Horizontally, G is $2a\cos\alpha - d = a\cos\alpha - b\sin\alpha$ nearer the wall than O, so the weight has moment $W(a\cos\alpha - b\sin\alpha)$, tending to turn the lamina towards the wall. At A, the force S acts along a line a height $2a\sin\alpha$ above O, and μS acts along the wall, a horizontal distance $2a\cos\alpha$ from O; both tend to turn the lamina away from the wall. So

$$W(a\cos\alpha - b\sin\alpha) = S\cdot 2a\sin\alpha + \mu S\cdot 2a\cos\alpha = \frac{2a\mu W(\sin\alpha + \mu\cos\alpha)}{1+\mu^2}. \qquad (*)$$

Ask yourself How does $\mu = \tan\lambda$ tidy up the right-hand side?

With $\mu = \tan\lambda$ we have $1 + \mu^2 = \sec^2\lambda$, so

$$\frac{\mu}{1+\mu^2} = \tan\lambda\cos^2\lambda = \sin\lambda\cos\lambda,$$
$$\sin\alpha + \mu\cos\alpha = \frac{\sin\alpha\cos\lambda + \cos\alpha\sin\lambda}{\cos\lambda} = \frac{\sin(\alpha+\lambda)}{\cos\lambda}.$$

So the right-hand side of $(*)$ is $2aW\sin\lambda\sin(\alpha+\lambda)$. Cancelling W and using $2\sin P\sin Q = \cos(P-Q) - \cos(P+Q)$ with $P = \alpha+\lambda$, $Q = \lambda$,

$$\begin{aligned} b\sin\alpha &= a\cos\alpha - 2a\sin\lambda\sin(\alpha+\lambda) \\ &= a\cos\alpha - a\big(\cos\alpha - \cos(\alpha+2\lambda)\big) \\ &= a\cos(2\lambda+\alpha), \end{aligned}$$

which is the required result.

The square lamina

If the lamina is square then $b = a$, and dividing by $a \neq 0$,

$$\cos(2\lambda+\alpha) = \sin\alpha = \cos\left(\tfrac{1}{2}\pi - \alpha\right).$$

So $2\lambda + \alpha = \pm\left(\frac{1}{2}\pi - \alpha\right) + 2k\pi$ for some integer k. We need to pick the right branch. Here $0 < \alpha < \frac{1}{2}\pi$ (the lamina leans), and $0 < \lambda < \frac{1}{2}\pi$ since $\mu > 0$, so $0 < 2\lambda + \alpha < \frac{3}{2}\pi$.

- The $+$ sign gives $2\lambda + \alpha = \frac{1}{2}\pi - \alpha + 2k\pi$; only $k = 0$ lands in $(0, \frac{3}{2}\pi)$ (since $\frac{1}{2}\pi - \alpha$ lies in $(0, \frac{1}{2}\pi)$), giving $\lambda = \frac{1}{4}\pi - \alpha$.
- The $-$ sign gives $2\lambda = -\frac{1}{2}\pi + 2k\pi$, i.e. $\lambda = -\frac{1}{4}\pi + k\pi$, and no such value lies in $(0, \frac{1}{2}\pi)$.

Hence $\lambda = \frac{1}{4}\pi - \alpha$, as required. (In particular equilibrium of this limiting kind needs $\alpha < \frac{1}{4}\pi$, which matches the need for G to lie on the wall side of O: $a\cos\alpha > a\sin\alpha$.)

Checks

- If $\mu = 0$ ($\lambda = 0$) the condition becomes $a\cos\alpha = b\sin\alpha$: G is directly above O, which is exactly when a lamina could balance on its corner with no help from the wall.
- For a square at $\alpha = 15°$ the result gives $\lambda = 30°$, so $\mu = 1/\sqrt{3} \approx 0.577$. Putting $a = b = 1$, $\alpha = \frac{\pi}{12}$, $\mu = 1/\sqrt{3}$ into $(*)$, both sides equal $W(\cos 15° - \sin 15°) \approx 0.707W$.
- Numerically, for 2000 random rectangles and angles we found μ from the force and moment equations by bisection and confirmed $a\cos(2\lambda+\alpha) = b\sin\alpha$ to within 10^{-14}.

Common slips

- Getting the directions of friction wrong: decide first which way each contact would slide, then oppose it.
- Using the distance $a\cos\alpha + b\sin\alpha$ from the *wall* as the moment arm about O; the arm about O is $a\cos\alpha - b\sin\alpha$.
- Forgetting that $\lambda = \frac{3}{4}\pi$ also solves the trigonometric equation; it is ruled out because $\mu > 0$ forces $0 < \lambda < \frac{1}{2}\pi$.

Beyond STEP

The angle λ with $\tan\lambda = \mu$ is the *angle of friction*: at limiting friction the total contact force makes angle λ with the normal. A neat alternative approach uses the fact that a rigid body in equilibrium under three forces needs them to be concurrent: the two total contact forces and the weight must meet in a point, which turns the problem into pure geometry. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops statics of rigid bodies and friction alongside the rest of Newtonian mechanics.

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